CHAPTER D

Curriculum coverage matrix

The expectations are rephrased and broken down into verifiable entries. “Covered” indicates a localised resource, not a mastery attested for the student. Not all of the optional extension items in the BO (Bulletin officiel) are compulsory. The seconde resources are maintained in R001–R030 and STATS.

ID	Expectation / type	Lesson, example	Exercise
BO001	Membership and inclusion (Content)	LOGIQUE, p. 130 ; LOGIQUE	LOG-EX, p. 130
BO002	Empty set, union, intersection and complement (Content)	LOGIQUE, p. 130 ; LOGIQUE	LOG-EX, p. 130
BO003	Sets of numbers and intervals (Content)	LOGIQUE, p. 130 ; LOGIQUE	LOG-EX, p. 130
BO004	Ordered pair and Cartesian product (Content)	LOGIQUE, p. 130 ; LOGIQUE	LOG-EX, p. 130
BO005	Cardinality of a finite set (Content)	LOGIQUE, p. 130 ; LOGIQUE	LOG-EX, p. 130
BO006	Connectives and/or (Skill)	LOGIQUE, p. 130 ; LOGIQUE	LOG-EX, p. 130
BO007	Counterexample (Skill)	LOGIQUE, p. 130 ; LOGIQUE	LOG-EX, p. 130
BO008	Implication and equivalence (Skill)	LOGIQUE, p. 130 ; LOGIQUE	LOG-EX, p. 130
BO009	Converse and contrapositive (Skill)	LOGIQUE, p. 130 ; LOGIQUE	LOG-EX, p. 130
BO010	Necessary and sufficient condition (Skill)	LOGIQUE, p. 130 ; LOGIQUE	LOG-EX, p. 130
BO011	Identity, equation, variable, unknown and parameter (Skill)	LOGIQUE, p. 130 ; LOGIQUE	LOG-EX, p. 130
BO012	Quantifiers and implicit quantification (Skill)	LOGIQUE, p. 130 ; LOGIQUE	LOG-EX, p. 130
BO013	Negation of quantified statements (Skill)	LOGIQUE, p. 130 ; LOGIQUE	LOG-EX, p. 130
BO014	Proof by cases (Skill)	LOGIQUE, p. 130 ; LOGIQUE	LOG-EX, p. 130
BO015	Proof by contradiction (Skill)	LOGIQUE, p. 130 ; LOGIQUE	LOG-EX, p. 130
BO016	Proof by contrapositive (Skill)	LOGIQUE, p. 130 ; LOGIQUE	LOG-EX, p. 130
BO017	Variables and assignment (Skill)	PYTHON, p. 131 ; PYTHON	TP, p. 132
BO018	Functions and return values (Skill)	PYTHON, p. 131 ; PYTHON	TP, p. 132
BO019	Bounded loops and unbounded loops with a stopping condition (Skill)	PYTHON, p. 131 ; PYTHON	TP, p. 132
BO020	Conditional statements (Skill)	PYTHON, p. 131 ; PYTHON	TP, p. 132
BO021	Lists defined in extension, successive appends and list comprehension (Skill)	PYTHON, p. 131 ; PYTHON	TP, p. 132
BO022	Manipulating, adding and removing elements and indices (Skill)	PYTHON, p. 131 ; PYTHON	TP, p. 132
BO023	Traversing a list and iterating over its elements (Skill)	PYTHON, p. 131 ; PYTHON	TP, p. 132

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BO024	Direct rate (Skill)	R006, p. 134 ; R006	R006, p. 134
BO025	Initial value (Skill)	R007, p. 135 ; R007	R007, p. 135
BO026	Percentage rate of change (Skill)	R008, p. 135 ; R008	R008, p. 135
BO027	Successive changes (Skill)	R009, p. 135 ; R009	R009, p. 135
BO028	Reverse change (Skill)	R010, p. 135 ; R010	R010, p. 135
BO029	Zero-product property (Skill)	R011, p. 135 ; R011	R011, p. 135
BO030	Sign of an affine expression (Skill)	R014, p. 136 ; R014	R014, p. 136
BO031	Sign of a factorised quadratic expression (Skill)	C02-U4, p. 17 ; C02-EX4	C02-E1, p. 19
BO032	Expanding, factorising and simplifying (Skill)	R012, p. 135 ; R012	R012, p. 135
BO033	Graphical equations and inequalities (Skill)	R078, p. 146 ; R078	R078, p. 146
BO034	Graphical sign and variations (Skill)	R078, p. 146 ; R078	R078, p. 146
BO035	Drawing a line from an equation or from a point and a gradient (Skill)	R016, p. 136 ; R016	R016, p. 136
BO036	Reading the equation of a line (Skill)	R015, p. 136 ; R015	R015, p. 136
BO037	Gradient from two points (Skill)	R015, p. 136 ; R015	R015, p. 136
BO038	Charts, histograms, bar charts, pie charts and box plots (Skill)	STATS, p. 132 ; STATS	STATS-EX, p. 132
BO039	Moving from data to a representation (Skill)	STATS, p. 132 ; STATS	STATS-EX, p. 132
BO040	Statistical measures (Skill)	STATS, p. 132 ; STATS	STATS-EX, p. 132
BO041	Conditional probabilities in a table or tree (Skill)	C05-U1, p. 54 ; C05-EX1	C05-E1, p. 57
BO042	Distinguishing intersection from two conditionings (Skill)	C05-U1, p. 54 ; C05-EX1	C05-E1, p. 57
BO043	Explicit, recursive, algorithmic modes and patterns (Content)	C01-U1, p. 3 ; C01-EX1	C01-E1, p. 6
BO044	Notation for the sequence and for a term (Content)	C01-U1, p. 3 ; C01-EX1	C01-E1, p. 6
BO045	Moving between natural language, algebra and graph (Skill)	C01-U1, p. 3 ; C01-EX1	C01-E5, p. 7
BO046	Modelling a pattern or a count (Skill)	C01-U1, p. 3 ; C01-EX1	C01-E5, p. 7
BO047	Calculating explicit, recursive or algorithmic terms (Skill)	C01-U1, p. 3 ; C01-EX1	C01-E5, p. 7
BO048	Arithmetic sequence, general term, constant increase, affine link (Content)	C01-U2, p. 3 ; C01-EX2	C01-E2, p. 6
BO049	Geometric sequence, general term, constant rate, exponential link (Content)	C01-U3, p. 4 ; C01-EX3	C01-E3, p. 6
BO050	General term of an arithmetic sequence (Proof)	C01-U2, p. 3 ; C01-EX2	C01-E8, p. 8
BO051	General term of a geometric sequence (Proof)	C01-U3, p. 4 ; C01-EX3	C01-E3, p. 6
BO052	Sum of the first integers (Proof)	C01-U4, p. 4 ; C01-EX4	C01-E4, p. 7
BO053	Sum of 1 to q to the power n (Proof)	C01-U4, p. 4 ; C01-EX4	C01-E4, p. 7
BO054	Calculating sums of consecutive terms (Skill)	C01-U4, p. 4 ; C01-EX4	C01-E4, p. 7
BO055	Determining the direction of variation (Skill)	C01-U3, p. 4 ; C01-EX3	C01-E6, p. 7
BO056	Modelling discrete linear or exponential growth (Skill)	C01-U3, p. 4 ; C01-EX3	C01-E6, p. 7
BO057	Intuitive limit: finite, infinite, or no limit (Content)	C01-U5, p. 5 ; C01-EX5	C01-E6, p. 7
BO058	Conjecturing a limit in simple cases (Skill)	C01-U5, p. 5 ; C01-EX5	C01-AP5, p. 5

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BO059	Factorised, expanded and canonical forms (Content)	C02-U1, p. 16 ; C02-EX1	C02-E1, p. 19
BO060	Roots and choice of form (Content)	C02-U1, p. 16 ; C02-EX1	C02-E1, p. 19
BO061	Sum and product of roots (Content)	C02-U3, p. 17 ; C02-EX3	C02-E3, p. 19
BO062	Discriminant, solving and possible factorisation (Content)	C02-U2, p. 16 ; C02-EX2	C02-E3, p. 19
BO063	Studying the sign of a factorised form (Skill)	C02-U4, p. 17 ; C02-EX4	C02-E5, p. 20
BO064	Choosing a form for an equation, inequality, optimisation or variations (Skill)	C02-U4, p. 17 ; C02-EX4	C02-E5, p. 20
BO065	Finding the polynomials vanishing at two distinct real numbers (Skill)	C02-U3, p. 17 ; C02-EX3	C02-E4, p. 19
BO066	Diversifying factorisation strategies (Skill)	C02-U3, p. 17 ; C02-EX3	C02-E4, p. 19
BO067	Solving the quadratic equation by proof by cases (Proof)	C02-U2, p. 16 ; C02-EX2	C02-E6, p. 20
BO068	Rate of change and secant lines (Content)	C04-U1, p. 42 ; C04-EX1	C04-E1, p. 44
BO069	Derivative at a point as the intuitive limit of the rate (Content)	C04-U2, p. 42 ; C04-EX2	C04-E2, p. 44
BO070	Tangent as the limit of secant lines, and gradient (Content)	C04-U3, p. 42 ; C04-EX3	C04-E5, p. 45
BO071	Tangent linear approximation (Content)	C04-U3, p. 42 ; C04-EX3	C04-E5, p. 45
BO072	Calculating a rate and the gradient of a secant (Skill)	C04-U1, p. 42 ; C04-EX1	C04-E1, p. 44
BO073	Interpreting gradient, instantaneous velocity and marginal cost (Skill)	C04-U2, p. 42 ; C04-EX2	C04-E3, p. 44
BO074	Determining a derivative at a point graphically (Skill)	C04-U3, p. 42 ; C04-EX3	C04-E4, p. 45
BO075	Constructing a tangent with a given gradient (Skill)	C04-U3, p. 42 ; C04-EX3	C04-E4, p. 45
BO076	Determining the equation of a tangent (Skill)	C04-U3, p. 42 ; C04-EX3	C04-E4, p. 45
BO077	Calculating an approximation of f(a+h) (Skill)	C04-U3, p. 42 ; C04-EX3	C04-E4, p. 45
BO078	Equation of the tangent (Proof)	C04-U3, p. 42 ; C04-EX3	C04-E2, p. 44
BO079	Square root not differentiable at 0 (Proof)	C04-U4, p. 43 ; C04-EX4	C04-AP4, p. 43
BO080	Differentiable function and derivative function (Content)	C06-U1, p. 67 ; C06-EX1	C06-E1, p. 70
BO081	Derivatives of the square, cube, inverse and square root (Content)	C06-U1, p. 67 ; C06-EX1	C06-E1, p. 70
BO082	Derivative of a positive or negative integer power (Content)	C06-U1, p. 67 ; C06-EX1	C06-E1, p. 70
BO083	Sum, product, reciprocal and quotient of differentiable functions (Content)	C06-U2, p. 67 ; C06-EX2	C06-E1, p. 70
BO084	Absolute value: differentiability at 0 (Content)	C04-U4, p. 43 ; C04-EX4	C04-AP4, p. 43
BO085	Calculating a derivative function using the operations (Skill)	C06-U2, p. 67 ; C06-EX2	C06-E1, p. 70
BO086	Derivative of the square function (Proof)	C06-U1, p. 67 ; C06-EX1	C04-E8, p. 46
BO087	Derivative of the inverse function (Proof)	C06-U1, p. 67 ; C06-EX1	C04-E8, p. 46

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BO088	Derivative of a product (Proof)	C06-U2, p. 67 ; C06-EX2	C06-E1, p. 70
BO089	Even and odd functions, graphical interpretations (Content)	C06-U4, p. 68 ; C06-EX4	C06-AP4, p. 69
BO090	Link between the sign of the derivative and variations on an interval (Content)	C06-U3, p. 67 ; C06-EX3	C06-E6, p. 71
BO091	Characterisation of constant functions (Content)	C06-U3, p. 67 ; C06-EX3	C06-E6, p. 71
BO092	Derivative at a point and tangent at an extremum (Content)	C06-U3, p. 67 ; C06-EX3	C06-E6, p. 71
BO093	Studying variations and extrema (Skill)	C06-U3, p. 67 ; C06-EX3	C06-E2, p. 70
BO094	Studying quadratics using differentiation (Skill)	C06-U3, p. 67 ; C06-EX3	C06-E2, p. 70
BO095	Solving an optimisation problem (Skill)	C06-U5, p. 69 ; C06-EX5	C06-E5, p. 71
BO096	Establishing an inequality (Skill)	C06-U4, p. 68 ; C06-EX4	C06-E3, p. 70
BO097	Studying the relative position of two curves (Skill)	C06-U4, p. 68 ; C06-EX4	C06-E3, p. 70
BO098	Definition by f' equal to f and f(0)=1 (Content)	C09-U1, p. 104 ; C09-EX1	C09-AP1, p. 104
BO099	Existence and uniqueness (accepted without proof) (Content)	C09-U1, p. 104 ; C09-EX1	C09-AP1, p. 104
BO100	The number e and power notation (Content)	C09-U1, p. 104 ; C09-EX1	C09-AP1, p. 104
BO101	Addition of arguments and product (Content)	C09-U2, p. 104 ; C09-EX2	C09-E1, p. 108
BO102	Inverse, quotient and powers (Content)	C09-U2, p. 104 ; C09-EX2	C09-E1, p. 108
BO103	Sign, direction of variation and curve (Content)	C09-U2, p. 104 ; C09-EX2	C09-E1, p. 108
BO104	Link with geometric sequences (Content)	C09-U4, p. 105 ; C09-EX4	C09-E8, p. 109
BO105	Transforming an exponential expression (Skill)	C09-U2, p. 104 ; C09-EX2	C09-E1, p. 108
BO106	Differentiating exp(at) (Skill)	C09-U3, p. 105 ; C09-EX3	C09-E3, p. 108
BO107	Representing exp(kt) and exp(-kt) for positive k (Skill)	C09-U4, p. 105 ; C09-EX4	C09-E4, p. 108
BO108	Modelling exponential growth or decay (Skill)	C09-U4, p. 105 ; C09-EX4	C09-E4, p. 108
BO109	Unit circle, arc and radian (Content)	C10-U1, p. 118 ; C10-EX1	C10-E1, p. 121
BO110	Wrapping the real line and image of a real number (Content)	C10-U2, p. 118 ; C10-EX2	C10-E2, p. 121
BO111	Sine, cosine, right-angled triangle and special values (Content)	C10-U3, p. 118 ; C10-EX3	C10-E3, p. 121
BO112	Placing a point on the circle (Skill)	C10-U2, p. 118 ; C10-EX2	C10-E2, p. 121
BO113	Reading associated angles and their sines and cosines (Skill)	C10-U4, p. 119 ; C10-EX4	C10-E3, p. 121
BO114	Cosine and sine of $\frac{\pi}{4}$ (Proof)	C10-U3, p. 118 ; C10-EX3	C10-RECH, p. 117
BO115	Cosine and sine of $\frac{\pi}{3}$ (Proof)	C10-U3, p. 118 ; C10-EX3	C10-RECH, p. 117
BO116	Dot product by projection and cosine (Content)	C03-U1, p. 29 ; C03-EX1	C03-E4, p. 32

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BO117	Characterisation of orthogonality (Content)	C03-U1, p. 29 ; C03-EX1	C03-E4, p. 32
BO118	Expression in an orthonormal basis and norm (Content)	C03-U2, p. 29 ; C03-EX2	C03-E1, p. 32
BO119	Coordinates via dot products with the basis vectors (Content)	C03-U2, p. 29 ; C03-EX2	C03-E1, p. 32
BO120	Bilinearity and symmetry (Content)	C03-U3, p. 30 ; C03-EX3	C03-E3, p. 32
BO121	Squares of the norms of a sum and a difference (Content)	C03-U3, p. 30 ; C03-EX3	C03-E3, p. 32
BO122	Law of cosines (Al-Kashi formula) (Content)	C03-U4, p. 30 ; C03-EX4	C03-E8, p. 33
BO123	Transformation of MA dot MB (Content)	C03-U5, p. 30 ; C03-EX5	C03-E5, p. 33
BO124	Calculating an angle and a length (Skill)	C03-U4, p. 30 ; C03-EX4	C03-E3, p. 32
BO125	Choosing projection, coordinates, norms and angle (Skill)	C03-U4, p. 30 ; C03-EX4	C03-E3, p. 32
BO126	Solving a geometric problem (Skill)	C03-U4, p. 30 ; C03-EX4	C03-E3, p. 32
BO127	Proving orthogonality (Skill)	C03-U2, p. 29 ; C03-EX2	C03-E2, p. 32
BO128	Law of cosines (Al-Kashi formula) via the dot product (Proof)	C03-U4, p. 30 ; C03-EX4	C03-E8, p. 33
BO129	Locus of MA dot MB equal to zero (Proof)	C03-U5, p. 30 ; C03-EX5	C03-RECH, p. 28
BO130	Normal vector and Cartesian equation (Content)	C07-U1, p. 80 ; C07-EX1	C07-E1, p. 82
BO131	Orthogonal projection onto a line (Content)	C07-U2, p. 80 ; C07-EX2	C07-E2, p. 82
BO132	Equation of a circle (Content)	C07-U3, p. 81 ; C07-EX3	C07-E3, p. 82
BO133	Line through a point with a given normal (Skill)	C07-U1, p. 80 ; C07-EX1	C07-E1, p. 82
BO134	Coordinates of the orthogonal projection (Skill)	C07-U2, p. 80 ; C07-EX2	C07-E2, p. 82
BO135	Equation from the centre and the radius (Skill)	C07-U3, p. 81 ; C07-EX3	C07-E3, p. 82
BO136	Recognising a circle, its centre and its radius (Skill)	C07-U3, p. 81 ; C07-EX3	C07-E3, p. 82
BO137	Using a coordinate system for a configuration (Skill)	C07-U4, p. 81 ; C07-EX4	C07-E4, p. 82
BO138	Independence of two events (Content)	C05-U3, p. 55 ; C05-EX3	C05-E3, p. 57
BO139	Partition of the sample space and total probability (Content)	C05-U2, p. 54 ; C05-EX2	C05-E2, p. 57
BO140	Sequence of two independent trials (Content)	C05-U4, p. 55 ; C05-EX4	C05-E5, p. 58
BO141	Repetition of identical independent Bernoulli trials for n at most 4 (Content)	C05-U4, p. 55 ; C05-EX4	C05-E5, p. 58
BO142	Calculating using total probability (Skill)	C05-U2, p. 54 ; C05-EX2	C05-E2, p. 57
BO143	Using or justifying independence (Skill)	C05-U3, p. 55 ; C05-EX3	C05-E3, p. 57
BO144	Tree or table for two independent trials (Skill)	C05-U4, p. 55 ; C05-EX4	C05-E5, p. 58
BO145	Tree and calculation for n at most 4 Bernoulli trials (Skill)	C05-U4, p. 55 ; C05-EX4	C05-E5, p. 58

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BO146	Random variable as a function on a finite sample space (Content)	C08-U1, p. 91 ; C08-EX1	C08-E1, p. 94
BO147	Probability distribution (Content)	C08-U1, p. 91 ; C08-EX1	C08-E1, p. 94
BO148	Expected value (Content)	C08-U2, p. 91 ; C08-EX2	C08-E5, p. 95
BO149	Linearity of expected value (Content)	C08-U2, p. 91 ; C08-EX2	C08-E5, p. 95
BO150	Variance and standard deviation (Content)	C08-U3, p. 92 ; C08-EX3	C08-E2, p. 94
BO151	König–Huygens formula (Content)	C08-U3, p. 92 ; C08-EX3	C08-E2, p. 94
BO152	Interpreting the events X=a and X less than or equal to a (Skill)	C08-U1, p. 91 ; C08-EX1	C08-E1, p. 94
BO153	Moving between natural language and symbols (Skill)	C08-U1, p. 91 ; C08-EX1	C08-E1, p. 94
BO154	Modelling with a random variable (Skill)	C08-U1, p. 91 ; C08-EX1	C08-E1, p. 94
BO155	Determining a distribution (Skill)	C08-U1, p. 91 ; C08-EX1	C08-E1, p. 94
BO156	Calculating expected value, variance and standard deviation (Skill)	C08-U3, p. 92 ; C08-EX3	C08-E2, p. 94
BO157	Using the expected value for a fair game (Skill)	C08-U2, p. 91 ; C08-EX2	C08-E3, p. 94
BO158	Simulating a random variable (Skill)	C08-U5, p. 93 ; C08-EX5	TP, p. 132
BO159	Writing the mean of a sample (Skill)	C08-U5, p. 93 ; C08-EX5	TP, p. 132
BO160	Comparing simulated mean and expected value (Skill)	C08-U5, p. 93 ; C08-EX5	TP, p. 132
BO161	Simulating N samples of size n and the frequency within 2 sigma over root n (Skill)	C08-U5, p. 93 ; C08-EX5	TP, p. 132
BO162	Terms, sums and thresholds (Algorithmic example)	C01-U5, p. 5 ; C01-EX5	TP, p. 132
BO163	Factorial, Fibonacci and Collatz (Algorithmic example)	PYTHON, p. 131 ; PYTHON	TP, p. 132
BO164	List of the gradients of secant lines (Algorithmic example)	C04-U1, p. 42 ; C04-EX1	TP, p. 132
BO167	Approximation of pi by Archimedes (Algorithmic example)	C10-U5, p. 120 ; C10-EX5	TP, p. 132
BO168	Monte Carlo method: parabola and disc (Algorithmic example)	C05-U5, p. 56 ; C05-EX5	TP, p. 132
BO169	Expected value, variance and standard deviation in Python (Algorithmic example)	C08-U3, p. 92 ; C08-EX3	TP, p. 132
BO170	Letter frequencies (Algorithmic example)	C08-E7, p. 95 ; C08-E7	TP, p. 132